PEPII

RF SYSTEM

BELLOW CAVITY PHASING PROCEDURE

PS-340-330-60-R1

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Station #:_____ Date:_____ Initl.:_____

BELLOW CAVITY PHASING PROCEDURE & DATA

1) Introduction

The phase relationship between 4 (HER) or 2 (LER) cavities of each station has been optimized at installation to bring the RF phase of each cavity in coincidence with beam arrival.
Some touch-up adjustment may be required if the balance between the cavity gap voltages of a station is not constant under high beam current conditions. Small adjustments in the length of the bellows of the waveguide splitting network can correct this imbalance. The procedure is described below:

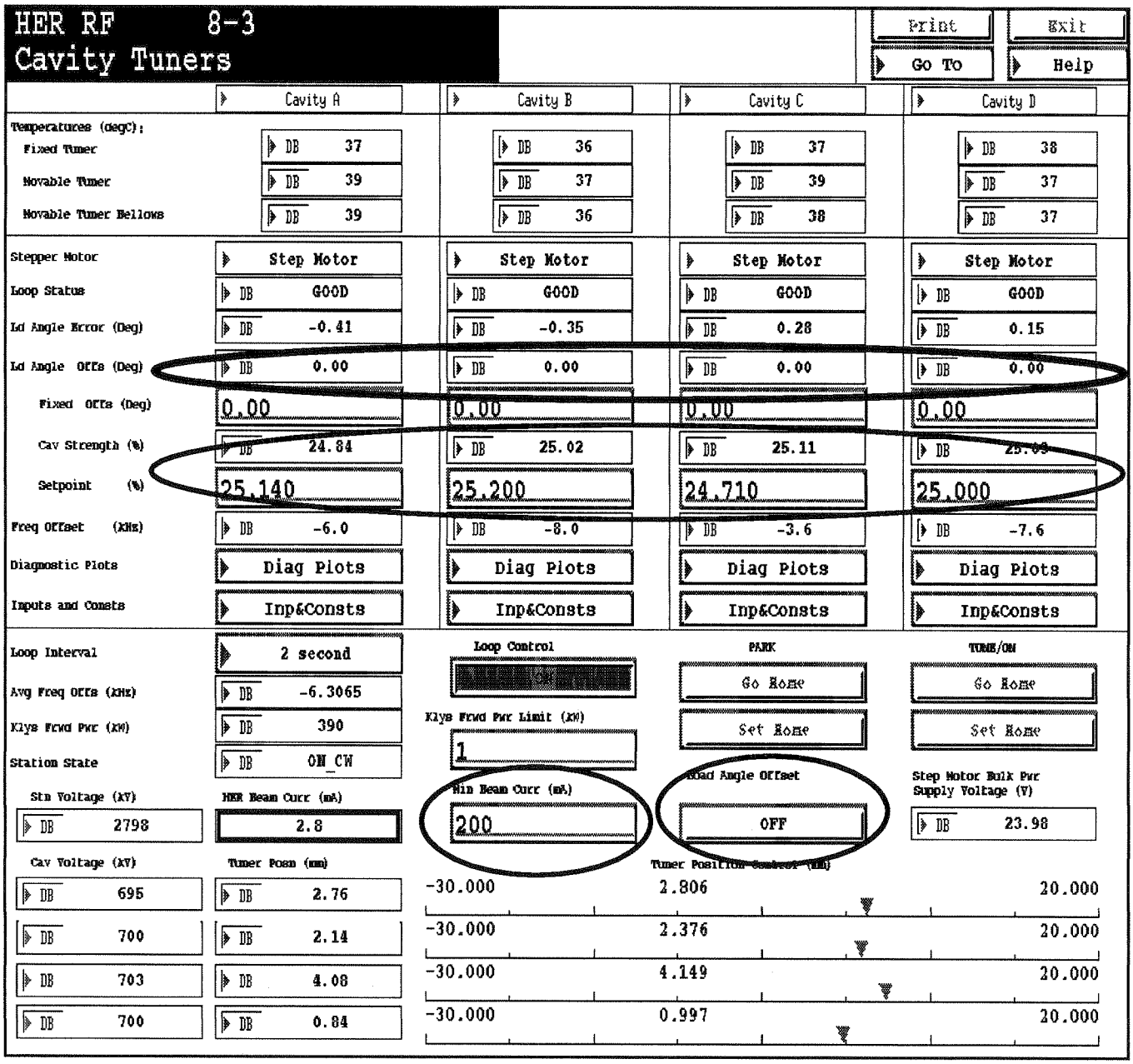
2) Measurement of Phase Error Procedure:

The **Load Angle Offset** routine is used to measure the magnitude of the phase offset of individual cavities of a station in reference to the beam.
The set-up and measurement is done using the **Cavity Tuners Panel**:
Initially with zero beam the **Cav Strength (%)** numbers need to be transferred to the **Setpoint (%)** data input box to set the idling condition of the cavities of a station.
Next with a stored beam larger then the **Min Beam Curr (mA)** set on the same panel the **Load Angle Offset** routine can be turned **ON** to activate the Load Angle Offset Loop which moves the tuners to bring the Cav Strength (%) reading with beam back to the idling cavity strength. The phase offsets for each cavity can then be read as **Ld Angle Offset (Deg)** on the Cavity Tuner Panel. The Load Angle Offset Loop is an iterative loop which will take a few minutes to stabilize (The Ld Angle Error box will go yellow everytime a tuner is asked to move).

Record the Angle Offsets after the loop has stabilized in the Data table under 4).

Note: The **Load Angle Offset** routine has to be turned **Off** again after this measurement because it interferes with the Magic Detuning required for high beam current operation.

Cavity Tuners EPICS Panel:



Station #:_____

4) Data:

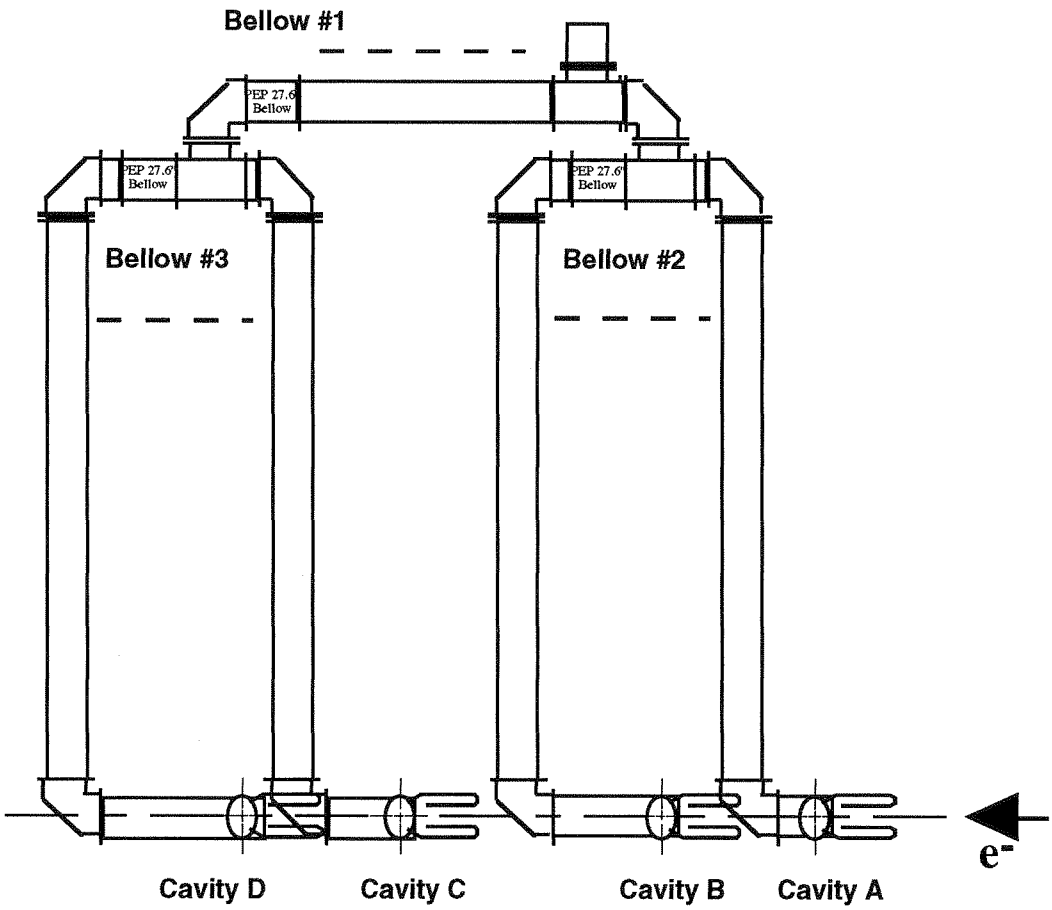
Date				
Beam Curr.				
	Ld Angle Offset (Deg)	Delta Phase = Ph_N - Ph_A	# bellow	Δ length (inch) = - 0.085 * ΔPh
cav. A (Ref)		0		0
cav. B			#2	
cav. C			#1	
cav. D			#3	

Calculate: Delta Phase = P_N - P_A.
If Delta Phase is **positive: make bellow shorter**
If Delta Phase is negative: make bellow longer
Bellow change is 0.085 inch/degree.
Note: Adjustment of bellow #1 for cavity C also shifts cavity D and needs to be compensated for by counter adjusting bellow #3.

Repeat if necessary:

Date				
Beam Curr.				
	Ld Angle Offset (Deg)	Delta Phase = Ph_N - Ph_A	# bellow	Δ length (inch) = - 0.085 * ΔPh
cav. A (Ref)		0		0
cav. B			#2	
cav. C			#1	
cav. D			#3	

Station #:_____



Date	Bellow #2	Bellow #1	Bellow #3